www.tntextbooks.in

respectively. They are kept on a horizontal table parallel to each other with their north poles pointing towards the south. They have a common magnetic equator and are separated by a distance of 20.0 cm. The value of the resultant horizontal magnetic induction at the mid-point O of the line joining their centers is (Horizontal components of Earth's magnetic induction is $3.6 \times 10^{-5}$ Wb $m^{-2}$ )

(NSEP 2000-2001)

(a) $3.60 \times 10^{-5}$ Wb $m^{-2}$

(b) $3.5 \times 10^{-5}$ Wb $m^{-2}$

(c) $2.56 \times 10^{-4}$ Wb $m^{-2}$

(d) $2.2 \times 10^{-4}$ Wb $m^{-2}$

**13.** The vertical component of Earth's magnetic field at a place is equal to the horizontal component. What is the value of angle of dip at this place?

(a) 30° (b) 45°

(c) 60° (d) 90°

**14.** A flat dielectric disc of radius R carries an excess charge on its surface. The surface charge density is σ. The disc rotates about an axis perpendicular to its plane passing through the center with angular velocity ω. Find the magnitude of the torque on the disc if it is placed in a uniform magnetic field whose strength is B which is directed perpendicular to the axis of rotation

(a) $\frac{1}{4}\sigma\omega\pi BR$ (b) $\frac{1}{4}\sigma\omega\pi BR^2$

(c) $\frac{1}{4}\sigma\omega\pi BR^3$ (d) $\frac{1}{4}\sigma\omega\pi BR^4$

**15.** A simple pendulum with charged bob is oscillating with time period T and let θ be the angular displacement. If the uniform magnetic field is switched ON in a direction perpendicular to the plane of oscillation then

(a) time period will decrease but θ will remain constant

(b) time period remain constant but θ will decrease

(c) both T and θ will remain the same

(d) both T and θ will decrease

## Answers

| | | | | |
|---|---|---|---|---|
| **1)** a | **2)** d | **3)** c | **4)** b | **5)** b |
| **6)** a | **7)** b | **8)** a | **9)** b | **10)** c |
| **11)** b | **12)** c | **13)** b | **14)** d | **15)** c |

## II Short answer questions:

**1.** What is meant by magnetic induction?

**2.** Define magnetic flux.

**3.** Define magnetic dipole moment.

**4.** State Coulomb's inverse law.

**5.** What is magnetic susceptibility?

**6.** State Biot-Savart's law.

**7.** What is magnetic permeability?

**8.** State Ampere's circuital law.

**9.** Compare dia, para and ferro-magnetism.

**10.** What is meant by hysteresis?



UNIT-3(XII-Physics_Vol-1).indd 201 04-03-2019 10:36:11